# Takuya AOYAMA


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## Personal

Born on May 7, 1987.

Japan Citizen.

## Education

2010 B.S. (Science) Science, Osaka University, Japan.

2012 M.A. (Science) Engineering science, Osaka University, Japan.

2015 Ph.D. (Science) Engineering science, Osaka University, Japan.

## Employment

Tohoku University 2015/4/1 - 2024/3/31.

Hiroshima University 2024/4/1-

## Committee service

Affiliated academic society: Physical Society of Japan, Japan Society of High Pressure Science and Technology

Editorial committee of the review of high pressure science and technology, 2020-2023.

## Funding Information

KAKENHI (See https://nrid.nii.ac.jp/en/nrid/1000080757261/)

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Murata science foundation (R06)

# Publications

## *Journal Articles*

1. R. Ohtani, Y. Iwai, S. Yamamoto, S. Maeda, K. Kume, Y. Masubuchi, T. Aoyama, Y. Tsuji, A. Kuwabara, S. I. Kawaguchi, H, Kadobayashi, N. Hirao, B. Le Ouay, M. Ohba, Giant Negative Linear Compression along the Stacking Direction of a Highly Compressive Two-Dimensional Coordination Polymer, Chem. Mater., **38**, 8930 (2026).

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3. R. Oishi, K. Umeo, T. Aoyama, T. Onimaru, and K. Kobayashi, Piezomagnetic Effect of a Rare-earth-based Altermagnet $TbPt_6Al_3$, J. Phys. Soc. Jpn., **95**, 063701 (2026).

4. T. Aoyama and K. Ohgushi, Magnetotransport Properties of Iron-Based Ladder Compounds $BaFe_2S_3$ and $BaFe_2Se_3$ under High Pressure, J. Phys. Soc. Jpn., **95**, 054702 (2026).

5. S. Ikeda, H. Tamura, S. Kawachi, H. Kadobayashi, S. I. Kawaguchi, S. Imaizumi, T. Aoyama, K. Ohgushi, and H. Kobayashi, Block-type antiferromagnetic state in $BaFe_2Se_3$ studied by $^{57}$Fe nuclear forward scattering under pressure, J. Phys.: Condens. Matter., **38** 145802 (2026).

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8. D. Takegami, T. Aoyama, T. Okauchi, T. Yamaguchi, S. Tippireddy, S. Agrestini, M. García-Fernández, T. Mizokawa, K. Ohgushi, Ke-Jin Zhou, J. Chaloupka, J. Kuneš, A. Hariki, and H. Suzuki, Circular Dichroism in Resonant Inelastic X-Ray Scattering: Probing Altermagnetic Domains in MnTe, Phys. Rev. Lett., **135**, 196502 (2025).

9. K. Kawaguchi, Y. Imai, T. Aoyama, S. I. Kawaguchi, and K. Ohgushi, Structural and Electronic Properties of Iron-based Ladder Compounds $BaFe_2(S_{1-x}Te_x)_3$, J. Phys. Soc. Jpn., **94**, 124701 (2025).

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14. F. Sato, T. Aoyama, S. Kawaguchi, H. Gotou, Y. Imai, and K. Ohgushi, Structural and Electronic Properties of $RuX_3$ ($X$ = Br and I) under High Pressure, J. Phys. Soc. Jpn., **93**, 093601 (2024).
15. H.-C. Wu, T. Aoyama, D. Morikawa, D. Okuyama, K. Nawa, W.-t. Chen , C.-H. Lu, T.-W. Yen, S.-M. Huang, S. Calder, S. Torii, K. Ohgushi, M. Terauchi, and T. J. Sato, Observation of Thermally Induced Piezomagnetic Switching in $Cu_2OSeO_3$ Polymorph Synthesized under High-Pressure, Adv. Phys. Res., 2400054 (2024).
16. T. Aoyama and K. Ohgushi, Piezomagnetic properties in altermagnetic MnTe, Phys. Rev. Materials, **8**, L041402 (2024).
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20. T. Zhu, X.-Z. Lu, T. Aoyama, K. Fujita, Y. Nambu, T. Saito, H. Takatsu, T. Kawasaki, T. Terauchi, S. Kurosawa, A. Yamaji, H.-B. Li, C. Tassel, K. Ohgushi, J. M. Rondinelli, and H. Kageyama, Thermal multiferroics in all-inorganic quasi-two-dimensional halide perovskites, Nat. Mater., **23**, 182 (2024).
21. T. Aoyama, R. Nojima, T. Yamauchi, Y. Imai, and K.Ohgushi, Pressure-induced switching of electronic nematicity in the iron-based ladder materials $BaFe_2(S_{1-x}Se_x)_3$ ($x$ = 0-1), *Phys. Rev.* B, **108**, 094507 (2023).
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25. H. Takahashi, R. Kikuchi, C. Kawashima, S. Imaizumi, T. Aoyama, and K. Ohgushi, Pressure-Induced Superconductivity in Iron-Based Spin-Ladder Compound $BaFe_{2+\delta}(S_{1-x}Se_x)_3$, Materials **15**, 1401 (2022).
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31. Y. Imai, K. Sasaki, T. Aoyama, K. Shirasaki, T. Yamamura, and K. Ohgushi, High-pressure synthesis of heavily hole-doped cuprates $Mg_{1-x}Li_xCu_2O_3$ with quasi-one-dimensional structure, *Phys. Rev.* B **101**, 245112 (2020).

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43. T. Aoyama, A. Iyama, K. Shimizu and T. Kimura, Multiferroicity in orthorhombic $R$MnO$_3$ ($R$ = Dy, Tb and Gd) under high pressure, *Phys. Rev.* B **91**, 081107(R) (2015).
44. T. Aoyama, K. Yamauchi, A.Iyama, S. Picozzi, K. Shimizu and T. Kimura, Giant spin-driven ferroelectric polarization in $TbMnO_3$ under high pressure, *Nat Commn.* **5** 4927 (2014).
45. T. Honda, T. Aoyama, J. S. White, Th. Strassle, L. Keller, M. Kenzelmann, F. Honda, A. Miyake, K. Shimizu, Y. Wakabayashi and T. Kimura, Pressure effect on magnetism and multiferroicity in $Mn_2GeO_4$, *Phys. Rev.* B **89**, 104405 (2014).
46. T. Aoyama, A. Miyake, T. Kagayama, K. Shimizu and T. Kimura, Pressure effects on magnetoelectric properties of a multiferroic triangular lattice antiferromagnet $CuCrO_2$, *Phys. Rev.* B **87**, 094401 (2013).

## *Proceedings*

1. T. Amano, Y. Kawakami, H. Itoh, T. Aoyama, Y. Imai, K. Ohgushi, Y. Nakamura, H. Kishida, K. Yonemitsu, and S. Iwai Ultrafast magnetization driven by spiral current in Kitaev spin liquid $\alpha$-$RuCl_3$, The International Conference on Ultrafast Phenomena (UP) 2022, paper Tu2A.2. (2022).
2. T. Aoyama, A. Iyama, K. Shimizu and T. Kimura, Magnetic-field-induced ferroelectric polarization flop under pressure in $TbMnO_3$, Journal of Physics: Conference Series **592**, 012118 (2015).
3. T. Aoyama, A. Miyake, K. Shimizu and T. Kimura, Dielectric and AC-calorimetry Measurements of $SmMnO_3$ under High Pressure, *J. Phys. Soc. Jpn.* **81** SB036 (2012).
4. M. Soda, T. Aoyama, Y. Wakabayashi and K. Hirota Magnetism in Trianglar Lattice System $LuFeCoO_4$ having Relaxor Property Journal of Physics: Conference Series **320**, 012084 (2011).

## *Others*

1. 青山 拓也, 三宅 厚志, 清水 克哉, 木村 剛, ダイヤモンドアンビルセルを用いた誘電率・交流比熱の同時測定, 高圧力の科学と技術, 2015 年 25 巻 4 号 298-307.
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3. 青山 拓也, 大串 研也, 梯子型鉄系化合物 $BaFe_2(S_{1-x}Se_x)_3$ におけるネマティック秩序, 固体物理 58, 283 (2023).
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5. T. Aoyama, JPSJ News Comments **22**, 17 (2025).